

2.11 Logic-Based Reasoning and Inference

Logic-based reasoning and inference are ways of thinking that help us draw conclusions from known facts using rules of logic. This improves critical thinking and is essential for solving complex real-world problems.

Logical Inference

Logical inference is the process of deriving new facts from known information.

Example:

- All humans are mortal
- Socrates is a human $\rightarrow$ Socrates is mortal

Inference ensures conclusions are true if the premises are true. It is used in mathematics, computer science, and artificial intelligence.

Deduction Using Logical Rules

Deduction is a type of reasoning that starts from general facts to reach a specific conclusion.

Example:

- If it rains, the ground gets wet
- It is raining $\rightarrow$ The ground is wet

Deduction is reliable because the conclusion is guaranteed to be true if the premises are true. Computers use deduction to solve problems and answer questions.

Short Questions

1. What is logic-based reasoning?
2. What is logical inference?
3. Give one example of logical inference.
4. What is deduction?
5. How does deduction work? Explain with an example.
6. Why is logical inference important?
7. Where is logical inference used?